

outlined: beyond the 32×/0.5× colour range ▽: slower than the column to its left corner: same configuration, auto-vectorizer on †: auto-vectorizer off in Vanilla Gibbon too

Real-world	ColorOctree	1×	1.08× 1.07×	▽ 0.718× 0.732×	0.826× 0.816×	1.15× 1.13×	1.17× 1.14×	1.17× 1.14×
	Compiler	1×	3.5× 3.5×	4.26× 4.49×	▽ 4.14× 4.17×	9.01× 9.43×	9.3× 9.38×	9.3× 9.38×
	DBQuery	1×	1.22× 1.24×	▽ 1.18× 1.17×	1.21× 1.21×	1.96× 1.97×	1.97× 1.97×	1.97× 1.97×
	DecisionTree	1×	1.83× 1.86×	2.76× 2.76×	2.77× 2.76×	2.77× 2.75×	▽ 2.76× 2.75×	2.76× 2.75×
	DecisionTreeClassify	1×	1.06× 1.04×	▽ 1.02× 0.973×	1.02× 0.98×	1.02× 0.978×	1.02× 0.979×	1.02× 0.979×
	DomTree	1×	1.3× 1.3×	1.32× 1.33×	▽ 1.25× 1.28×	1.33× 1.34×	1.34× 1.36×	1.34× 1.36×
	KDTree	1×	1.22× 1.22×	1.31× 1.21×	▽ 1.3× 1.2×	1.31× 1.2×	1.31× 1.2×	1.31× 1.2×
	ObjectGraph	1×	1.6× 1.61×	1.78× 1.59×	2.01× 2.06×	2.43× 2.43×	2.43× 2.41×	2.43× 2.41×
	OctTree	1×	1.04× 1.04×	▽ 0.934× 0.935×	1.01× 1.03×	1.42× 1.44×	1.42× 1.43×	1.42× 1.43×
	PiecewiseFunctions	1×	1.33× 1.34×	1.38× 1.39×	1.58× 1.6×	2.45× 2.43×	2.45× 2.41×	2.45× 2.41×
	Trie	1×	1.26× 1.27×	1.29× 1.27×	▽ 1.27× 1.27×	2.32× 2.32×	2.34× 2.35×	2.34× 2.35×
Synthetic	Add1TreeInt8	1×	1.16× 1.04×	1.18× 1.21×	1.74× 2.07×	2.12× 2.2×	2.17× 2.35×	2.17× 2.35×
	Add1TreeInt16	1×	1.3× 1.29×	▽ 1.19× 1.21×	1.76× 1.96×	2.07× 2.13×	2.18× 2.02×	2.18× 2.02×
	Add1TreeInt32	1×	1.15× 1.23×	1.21× 1.2×	1.65× 1.72×	1.91× 1.91×	▽ 1.9× 1.88×	1.9× 1.88×
	Add1TreeInt64	1×	1.14× 1.21×	1.15× 1.13×	1.48× 1.56×	1.64× 1.67×	1.64× 1.65×	1.64× 1.65×
	LinearListReduction	1×	2.23× 2.26×	▽ 1.99× 2.12×	▽ 1.95× 1.97×	3.71× 3.82×	3.89× 4.04×	3.89× 4.04×
	MonoTree	1×	1.64× 1.67×	1.66× 1.65×	2.06× 2.04×	2.22× 2.23×	2.24× 2.25×	2.24× 2.25×
	TernaryTree	1×	1.12× 1.13×	▽ 1.11× 1.09×	1.31× 1.28×	1.36× 1.33×	▽ 1.32× 1.34×	1.32× 1.34×
Arithmetic intensity	ArithmeticIntensityInt8 †	1× 1.17×	▽ 0.962× 1.22×	1.03× 1.26×	1.1× 3.32×	1.14× 3.55×	3.66× 3.56×	3.66× 3.56×
	ArithmeticIntensityInt16 †	1× 1.99×	1.05× 2.18×	▽ 1.03× 2.25×	1.08× 3.68×	1.13× 4.12×	3.85× 3.95×	3.85× 3.95×
	ArithmeticIntensityInt32 †	1× 1.31×	1.03× 1.37×	1.05× 1.47×	1.11× 2.41×	1.16× 2.48×	2.61× 2.58×	2.61× 2.58×
	ArithmeticIntensityInt64 †	1× 0.968×	▽ 0.993× 1.05×	1.03× 1.03×	1.04× 1.05×	1.07× 1.07×	1.07× 1.07×	1.07× 1.07×
		Vanilla Gibbon (AoS)	Mutable cursors (AoS)	SoA layout	Loopification	Buffer sharing	Gibbon SIMD	Total